



1.4 Fetal Heart Tracings

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O2PUS sheet music

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fetal heart tracings = tool to assess fetal oxygenation, aids in assessment of fetal well being

and labor/delivery management. Top section = fetal heart rate. Bottom section = maternal

contractions

2. FHR monitor belt and microphone next to teacher's head

→ fetal heart rate monitored via belt and scalp electrodes (external or internal monitors)

3. Blue-faced kid playing the recorder

→ fetal response pathway = hypoxemia (low blood O₂) → hypoxia (low tissue O₂) → metabolic acidosis (overall acidotic state) → metabolic acidemia (low blood pH)

4. Blue tissue box

→ fetal response pathway = hypoxemia (low blood O₂) → hypoxia (low tissue O₂) → metabolic acidosis (overall acidotic state) → metabolic acidemia (low blood pH)

5. Acidic pink lemonade

→ fetal response pathway = hypoxemia (low blood O₂) → hypoxia (low tissue O₂) → metabolic acidosis (overall acidotic state) → metabolic acidemia (low blood pH)

6

Umbilical cord straw

6

umbilical cord blood pH = used to assess level of acidemia in fetus, often used as

quantitative evidence for/against intrapartum hypoxic injury

7. Brain hat flying off of blue-faced kid

→ consequence of prolonged hypoxia → neurological hypoxic ischemic injury

8. Bass section

→ components of fetal heart tracings = baseline variability, accelerations, decelerations

9. Middle bass w/ tempo 110-160bpm music

→ normal fetal heart rate = 110-160bpm

10. Right bass w/ frazzled player, tempo >160bpm music

→ fetal tachycardia >160bpm

11. Infection charm bracelet on frazzled bass player

→ causes of fetal tachycardia = infection, maternal fever, medications, thyrotoxicosis, fetal cardiac arrhythmias, placental abruption

12. Flame bandana on frazzled bass player

→ causes of fetal tachycardia = infection, maternal fever, medications, thyrotoxicosis, fetal cardiac arrhythmias, placental abruption

13. Pill bottle on tachycardia sheet music

→ causes of fetal tachycardia = infection, maternal fever, medications, thyrotoxicosis, fetal cardiac arrhythmias, placental abruption

14. Bowtie on frazzled bass player

→ causes of fetal tachycardia = infection, maternal fever, medications, thyrotoxicosis, fetal cardiac arrhythmias, placental abruption

15. Shaking hearts on frazzled bass

→ causes of fetal tachycardia = infection, maternal fever, medications, thyrotoxicosis, fetal cardiac arrhythmias, placental abruption

16

Placenta earring abrpting off of frazzled bass player

16

causes of fetal tachycardia = infection, maternal fever, medications, thyrotoxicosis, fetal

cardiac arrhythmias, placental abruption

17. Left bass w/ sleepy player, tempo <110bpm music

→ fetal bradycardia <110bpm

18. Blue bass of sleepy player

→ causes of fetal bradycardia = hypoxia, fetal AV block (assoc. w/ maternal SLE), beta-blockers, fetal hypothyroidism

19. Shield pin with heart on sleepy bass player

→ causes of fetal bradycardia = hypoxia, fetal AV block (assoc. w/ maternal SLE), beta-blockers, fetal hypothyroidism

20. Muted beta-bugle next to sleepy bass player

→ causes of fetal bradycardia = hypoxia, fetal AV block (assoc. w/ maternal SLE), beta-blockers, fetal hypothyroidism

21. Untied bowtie on sleepy bass player

→ causes of fetal bradycardia = hypoxia, fetal AV block (assoc. w/ maternal SLE), beta-blockers, fetal hypothyroidism

22. Violin section

→ fetal heart rate variability = average peak to trough fluctuation in baseline expressed in beats per minute, the “beat-to-beat” variation

23. Flat line connecting whOle notes

→ types of variability = absent (no fluctuation, flat line), minimal (amplitude <5bpm), moderate (amplitude 6-25bpm), marked (amplitude >25bpm)

24. “<5” eighth note w/ trill line

→ types of variability = absent (no fluctuation, flat line), minimal (amplitude <5bpm), moderate (amplitude 6-25bpm), marked (amplitude >25bpm)

25. Treble clef “6” - “25” bass clef/eighth note + moderate variability music

types of variability = absent (no fluctuation, flat line), minimal (amplitude <5bpm),

moderate (amplitude 6-25bpm), marked (amplitude >25bpm)

26. Star next to moderate variability music

→ features of a FHT that EXCLUDE fetal acidemia = moderate variability, accelerations

27. ">25" bass clef/eighth note + marked variability music

→ types of variability = absent (no fluctuation, flat line), minimal (amplitude <5bpm),

moderate (amplitude 6-25bpm), marked (amplitude >25bpm)

28. Axe player + foot on pedal

→ fetal heart rate accelerations = rise of FHR above baseline

29. "<32" alto/bass clef + 10/10 time signature

→ FHT acceleration if <32wks GA = 10x10 (peak of at least 10bpm above the baseline

sustained for at least 10 seconds)

30. ">32" alto/bass clef + 15/15 time signature

→ FHT acceleration if >32wks GA = 15x15 (peak of at least 15bpm above the baseline

sustained for at least 15 seconds)

31. Star on acceleration sheet music

→ features of a FHT that EXCLUDE fetal acidemia = moderate variability, accelerations

32. Drum section

→ fetal heart rate decelerations = dips below the baseline (types = early, variable, late)

33. Cymbal player w/ pink cymbals

→ maternal contractions = uterine contractions, measured by a tocometer

34. Drum next to top staff of FHT music

→ fetal heart tracing top section = fetal heart rate, fetal heart tracing bottom section =
maternal contractions

35. Cymbals next to the bottom staff of FHT music

fetal heart tracing top section = fetal heart rate, fetal heart tracing bottom section =

maternal contractions

36. Sun drum + early deceleration music

→ fetal heart rate early decelerations = a decrease in fetal heart rate from baseline, time from onset to nadir is at least 30 sec., nadir of the deceleration is at the same time as the peak of the contraction, return of the FHR to baseline by the end of the uterine contraction

37. Tight headband on sun drummer

→ cause of early decelerations in FHR = fetal head compression

38. Variety of drums + variable deceleration music

→ variable decelerations = an abrupt decrease in the fetal heart rate from baseline of at least 15bpm that lasts at least 15 sec., but the onset to nadir is LESS than 30 seconds and not more than 2 min

39. Variable drum player holding umbilical cord drumstick

→ cause of variable decelerations in FHR = umbilical cord compression

40. Moon shirt on drummer + late deceleration music

→ fetal heart rate late decelerations = a gradual decrease in fetal heart rate from baseline that is associated with a uterine contraction, onset to nadir of the deceleration is ≥ 30 sec., nadir of the deceleration is AFTER the peak of the contraction

41. Late drum player playing uterus drum turning blue

→ cause of late decelerations in FHR = uteroplacental insufficiency. If persistent insult → recurrent late decelerations (w/ $\geq 50\%$ contractions)

42. Fetal O2 poster

→ fetal oxygenation pathway = environment → maternal lungs → maternal heart → maternal blood vessels → uterus → placenta → umbilical cord → fetus

43. Today's menu VEAL-CHOP

→ decelerations mnemonic = VEAL-CHOP (variable-cord compression, early-head compression, accelerations-ok!, late-placental insufficiency)

44. Cymbal player's lapel microphone

measuring contraction strength = intrauterine pressure catheter (IUPC),
measured in

montevideo units (adequate ≥ 200 mvu)

45. Cymbal player's "5" feather on hat + "10" shoulder cords

→ tachysystole = >5 contractions in 10 minutes averaged over 30 minutes

46. Mountain of video tapes

→ measuring contraction strength = intrauterine pressure catheter (IUPC),
measured in

montevideo units (adequate ≥ 200 mvu)

47. "Qty: 200" of video tapes

→ measuring contraction strength = intrauterine pressure catheter (IUPC),
measured in

montevideo units (adequate ≥ 200 mvu)

48. "I: Beginner" movement of O2PUS

→ category I FHT = normal baseline (110-160bpm) + moderate variability, +/-
early decels +

accels, NO late/variable decels = normal fetal oxygenation status

49. "III: Advanced" movement of O2PUS

→ category III FHT = 1. Sinusoidal pattern OR 2. Absent variability + either a)
Bradycardia b)

Recurrent late decelerations c) Recurrent variable decelerations = abnormal
fetal

oxygenation status

50. Sine wave music in "III: Advanced" movement of O2PUS

→ category III FHT = 1. Sinusoidal pattern OR 2. Absent variability + either a)
Bradycardia b)

Recurrent late decelerations c) Recurrent variable decelerations = abnormal
fetal

oxygenation status

51. Flat line connecting whole notes + tempo <110 bpm + recurrent
deceleration music in "III:

Advanced" movement of O2PUS

→ category III FHT = 1. Sinusoidal pattern OR 2. Absent variability + either a)
Bradycardia b)

Recurrent late decelerations c) Recurrent variable decelerations = abnormal
fetal

oxygenation status

52. "II: Intermediate" movement of O2PUS

→ category II FHT = anything NOT category I or III = indeterminate fetal
oxygenation status

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Variable drummer dropping umbilical cord drumstick

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to resolve recurrent variable decelerations → relieve cord compression
(maternal

repositioning, amnioinfusion)

54. Variable drummer drinking water

→ to resolve recurrent variable decelerations → relieve cord compression
(maternal
repositioning, amnioinfusion)

55. Late deceleration drummer wearing O2 mask

→ to resolve recurrent late decelerations → improve oxygenation pathway
supplemental O2,
check maternal BP → give IVF, maternal repositioning to left lateral decubitus,
stop labor
induction agents (oxytocin)

56. Late deceleration drummer's ivy charm bracelet

→ to resolve recurrent late decelerations → improve oxygenation pathway
supplemental O2,
check maternal BP → give IVF, maternal repositioning to left lateral decubitus,
stop labor
induction agents (oxytocin)

57. Late deceleration drummer kicking stool to the left

→ to resolve recurrent late decelerations → improve oxygenation pathway
supplemental O2,
check maternal BP → give IVF, maternal repositioning to left lateral decubitus,
stop labor
induction agents (oxytocin)

58. "No more ox-bell!" sign

→ to resolve recurrent late declarations → improve oxygenation pathway
supplemental O2,
check maternal BP → give IVF, maternal repositioning to left lateral decubitus,
stop labor
induction agents (oxytocin)

59. "Do not disTURB" sign above tuba

→ to resolve a tetanic contraction → terbutaline (β 2-agonist)

60. Stork next to "III: Advanced" sheet music

→ if you see a category III FHT → DELIVER THE BABY!
1.4 featl heart tracings